
ERROR? if we are fitting Delta Phi_j there is a mistake- we fit Phi directly

5 件のメッセージ

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月18日 7:16

To: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Hello,

as you can see in the notes attached what we should be fitting in our calculations is $\Phi(E_\nu)$ up to $E_\nu = 1.8$ MeV and NOT Delta Phi as Ko said. Please check these two pages out and see if I am making a mistake in them.

The Φ_j should be an ansatz for $\Phi(E_\nu)$ for $E_\nu < 1.8$. So the values we should get should be for $\Phi(E_\nu)$, not for Delta Phi.

If we would be making our ansatz for Delta Phi there would be a missing contribution to the rate in each bin that we did not include (I think) thus our calculation would be wrong (see box on page 1).

Ko: it is imperative that you explain to us clearly what you did to say that we are fitting Delta Phi.

Should we meet again before we send the paper to be published. There is a mistake somewhere in what Ko did because otherwise we should get directly Phi, so I do not understand why the figures showed Delta Phi.

I hope the error is trivial and the calculation are correct. But without a clear explanation of what Ko did to get his figures for DeltaPhi instead of Phi I cannot know.

Best,
Graciela



GG Note; we are fitting Phi and not Delta Phi -Aug 17, 2026.pdf
1277K

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年8月18日 7:28

To: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

One more thing: since we should be obtaining Φ_j and not Delta Φ_j our comparison with the integrated band in Danny's old paper may also be wrong.

I do not understand how Ko got fits for only DeltaPhi_j when the formalism should get us Φ_j . What I also do not understand is how is that we only discover this issue because Danny looks carefully at the vertical scale in the figures.

Best,
Graciela
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Yasuda Koichiro <k.yasuda5160@gmail.com>

2026年8月18日 10:05

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

Dear Graciela

Thank you for the clarification. I will look into it and get back to you as soon as possible.

Best,
Ko

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[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Muping Chen <mpchen1864@gmail.com>, Danny Marfatia <dmarf8@hawaii.edu>

2026年8月18日 10:23

Thanks Ko,

You may send handwritten notes specifying the procedure in your code, similar to the way we explain in our paper (or in my notes) so one can understand very clearly what is in it. If you discussed your code with Muping and he agreed to what you did, please do discuss with him. Now you are in the same timezone so it should be easier for you to talk to each other.

Best,
Graciela

[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>
To: Graciela Gelmini <gelmini@physics.ucla.edu>
Cc: Yasuda Koichiro <k.yasuda5160@gmail.com>, Muping Chen <mpchen1864@gmail.com>

2026年8月18日 20:12

Graciela, I see your point. However, in the second line of the boxed equation (and the one preceding it) the coefficient "1.8 MeV" should be " $\Phi(1.8 \text{ MeV})$ ". Unfortunately $\Phi(1.8 \text{ MeV})$ is not tiny (see Fig. 1), so the error introduced may not be negligible, if in fact that's how Ko did the analysis.

If Ko did fit Φ (not $\Delta \Phi$), then it doesn't make sense that the best-fit and the theoretical curves were systematically lower in Fig. 7 than in Fig. 1. Ko, check if Fig. 1 is wrong, and the original Fig. 7 was correct.

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